

3. The diagram shows the different waves that make up the electromagnetic (em) spectrum.

Gamma	X-rays	Ultraviolet	Visible light	Infra-red	Microwaves	Radio waves
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(a) State in terms of **two** of their properties why they are arranged in this order from left to right. [2]

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(b) (i) State which em wave can be radiated as energy from the nuclei of radioactive materials. [1]

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(ii) State which em wave is used for satellite communications. [1]

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(c) The energy of X-rays ranges from $2.0 \times 10^{-14} \text{ J}$ to $2.1 \times 10^{-17} \text{ J}$. Which em wave would have an energy of $1.5 \times 10^{-12} \text{ J}$? [1]

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